



Validating Friction Modifier Performance Using Novel Under-Rail Ultrasonic Stress Measurement

M. Santoro - L.B. Foster

H. Brunskill - Peak to Peak Measurement Solutions

H. Shackleton – Peak to Peak Measurement Solutions



DALLAS, TX





Table of Contents

1. Overview of Lateral forces
2. L/V and Friction Management
3. Traditional Measurement Techniques
4. Ultrasonic Technology
5. Field Results
6. Nex Steps/Further Work
7. Conclusions





Lateral Force Measurement Review

Lateral forces are critical to railway operations because they directly impact safety, vehicle stability, and track structural integrity. Key reasons for the importance of lateral forces include:

- **Curving Performance:** These forces act as a guiding force to keep the vehicle safely navigating curves, but excessive force can result in excessive curving resistance and potential damage
- **Derailment Safety:** Lateral forces are used to calculate the Nadal Ratio (L/V ratio), which is the primary criterion for predicting the risk of wheel climb and are key in force related derailments.
- **Track Stability and Alignment:** Large lateral loads can cause severe stress on the track structure, leading to instability or buckling, placing a greater requiring on ballast resistance and maintenance.
- **Component Wear and Maintenance:** Excessive lateral forces lead to accelerated wear of insulators, fastening systems, wheel flanges, and rails, decreasing the lifespan of these components.
- **Vehicle Stability (Hunting):** As trains reach high speeds, conical wheels naturally follow a sinusoidal movement (hunting), causing lateral forces that can threaten stability and cause passenger discomfort.

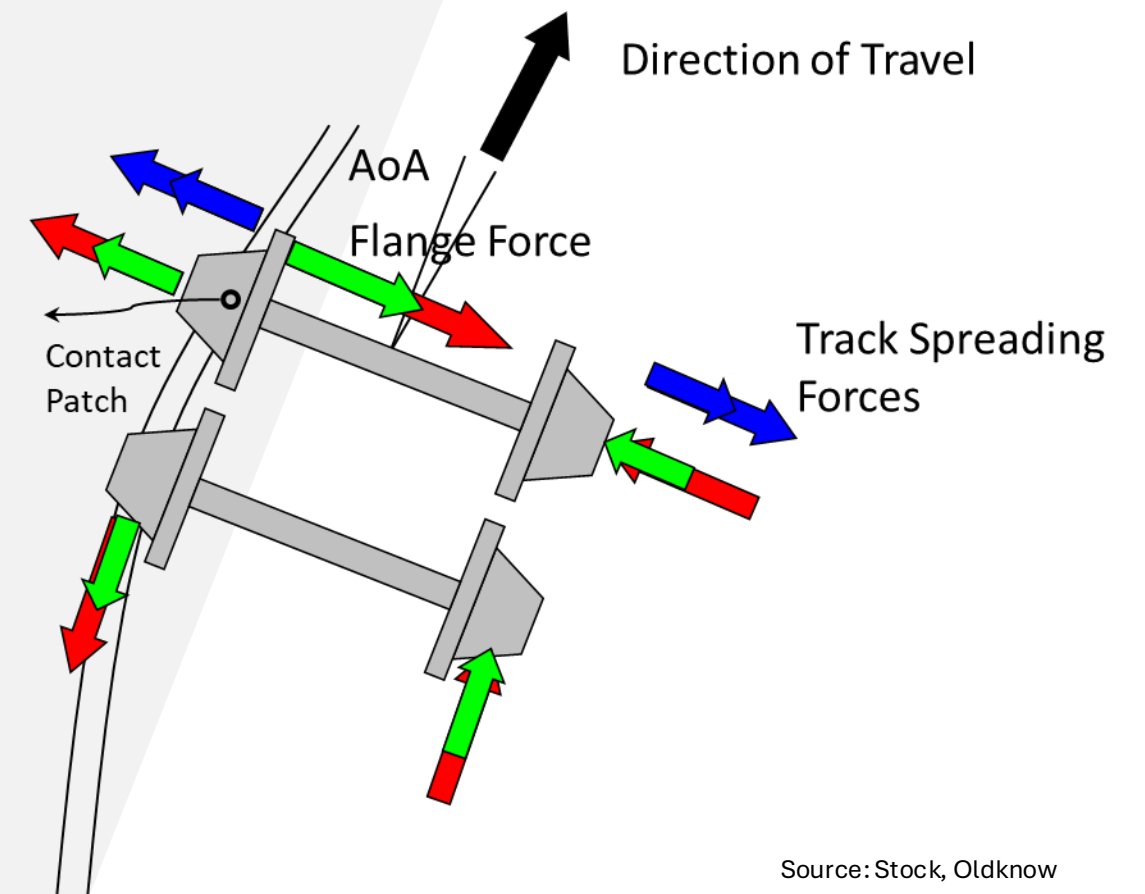
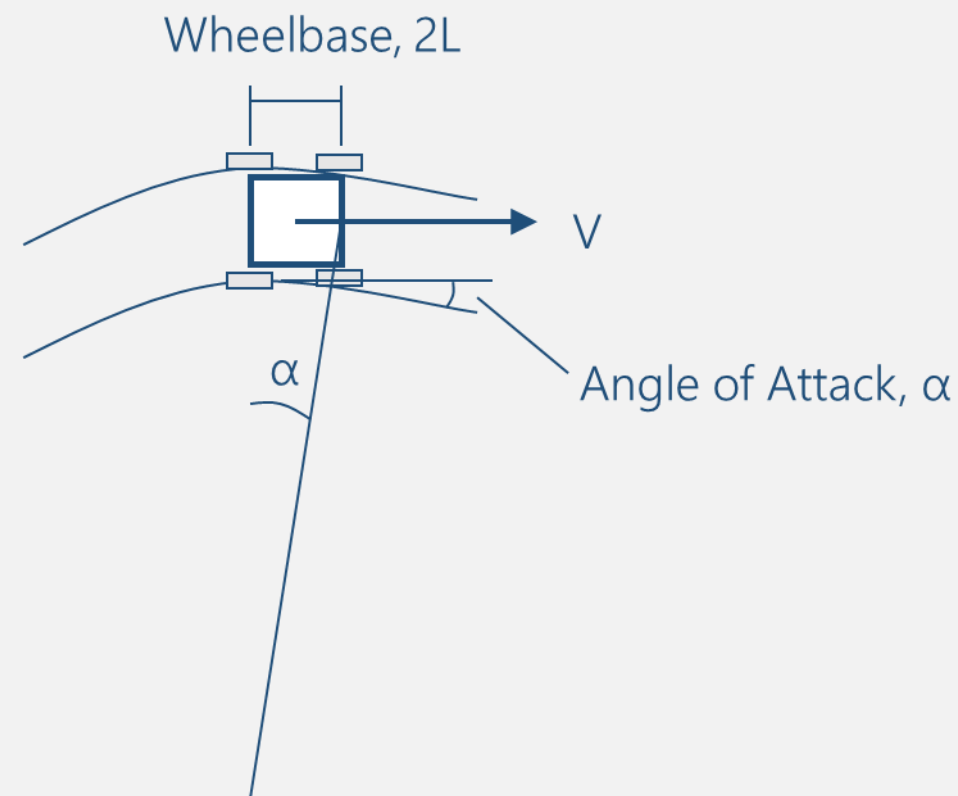
Friction modifiers have been proven to reduce L/V, reducing the stress on the track and providing benefits in all these key areas.

Lateral Force Measurement and Friction Modifiers

- Under saturated creep conditions, low and high rail lateral forces are primarily related to the coefficient of friction on the low rail

Low rail lateral force = Low rail **CoF** x Vertical force

High rail lateral force = Low rail **CoF** x Vertical force + some contribution from longitudinal forces



Source: Stock, Oldknow

- Top of rail friction modifiers are designed to control wheel tread/top of rail friction at an intermediate level
- Due to the thin film technology and distance traveled, friction measurements of the rails are not reliable
- The most common method of validation of product performance has been the use of L/V sites



L/V Data with TOR FM

- Lateral force measurements began in the early 2000s with tests at BNSF, CSX and CN.
- In the following years, major studies were carried out by TTCI at UP Mojave and by CP on the Western Canadian corridor.
- Results generated a 15-45% reduction in forces depending on high/low rail and general conditions. The usual average reduction when both rails are combined is 25%
- Many of the trial results have been presented/published at WRI, AREMA, IHHA, TTCI

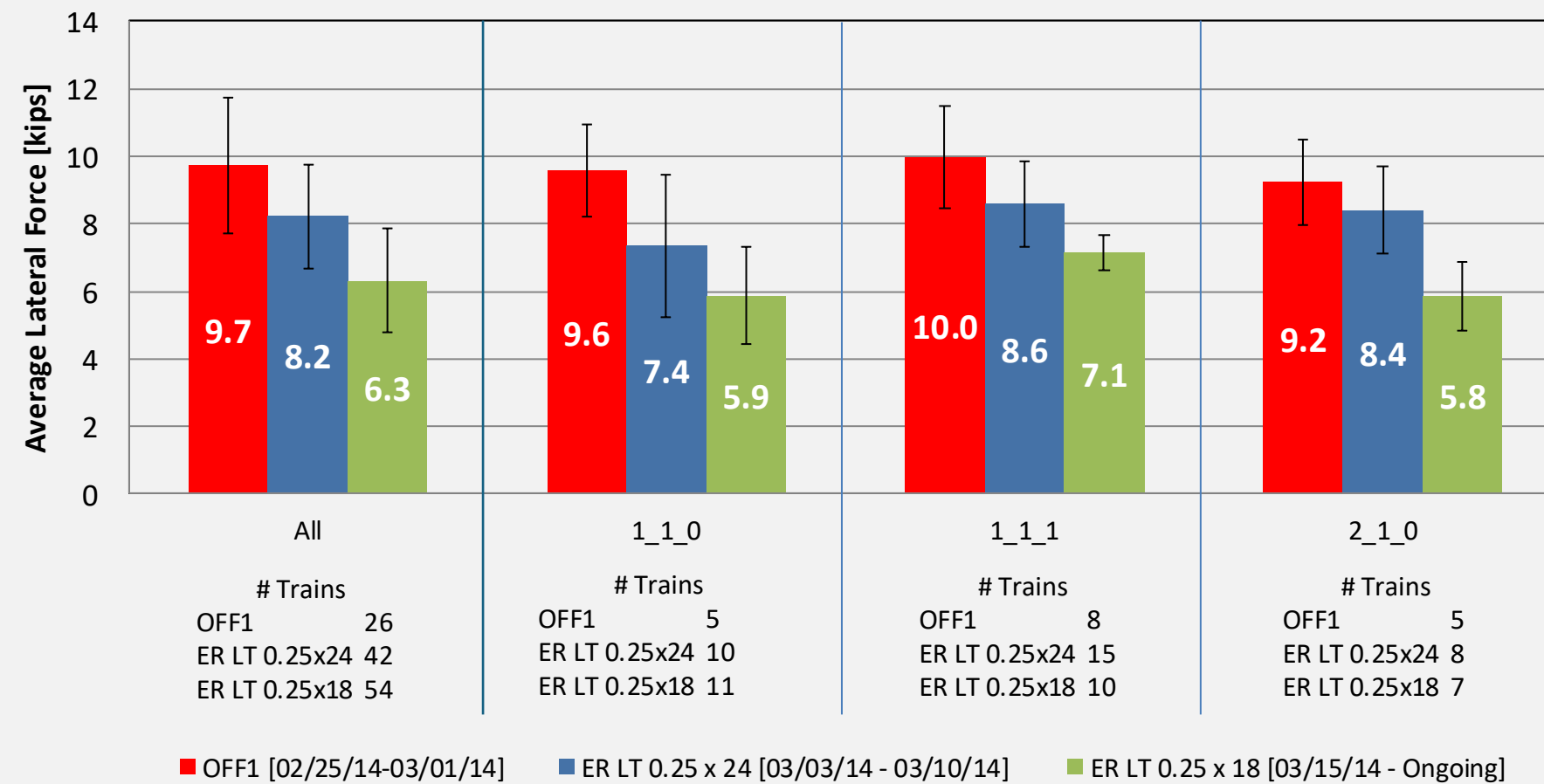




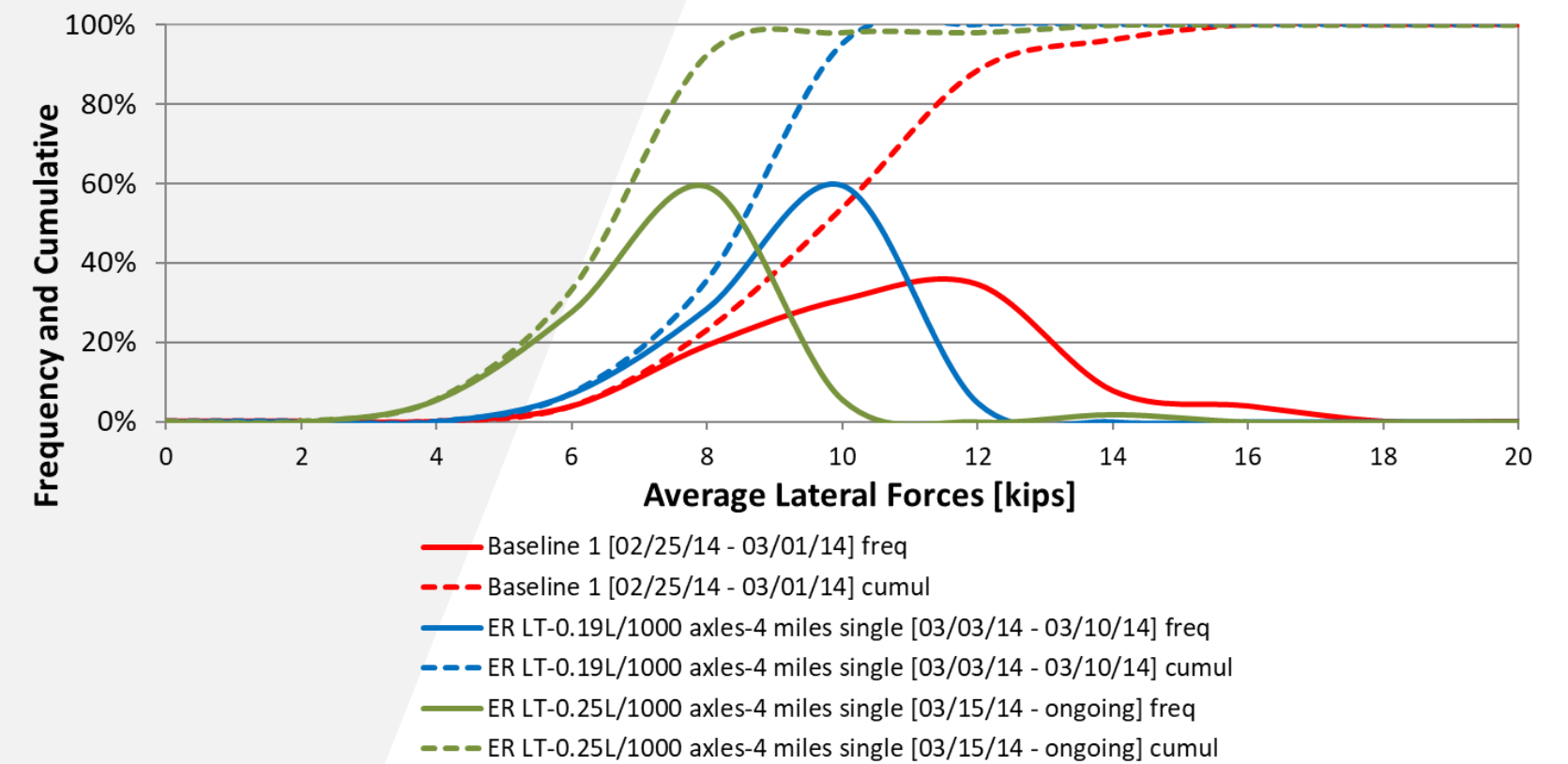
Friction Modifiers Results

- Class 1 trial results [high/low rail in various conditions]
- TOR + GF compared to GF only
- Shown as average values or distribution plot

ER LT Performance on Rolling Grade, 4.1 Miles Carry
MP 27.3, 7°30' RH, Main Track, Avg. Lat. Forces ISI Crib 1, Westbound, High Rail, Leading
Axles, Wheel Load > 25+ kips 24-28 mph speeds and ≥ 350 axles



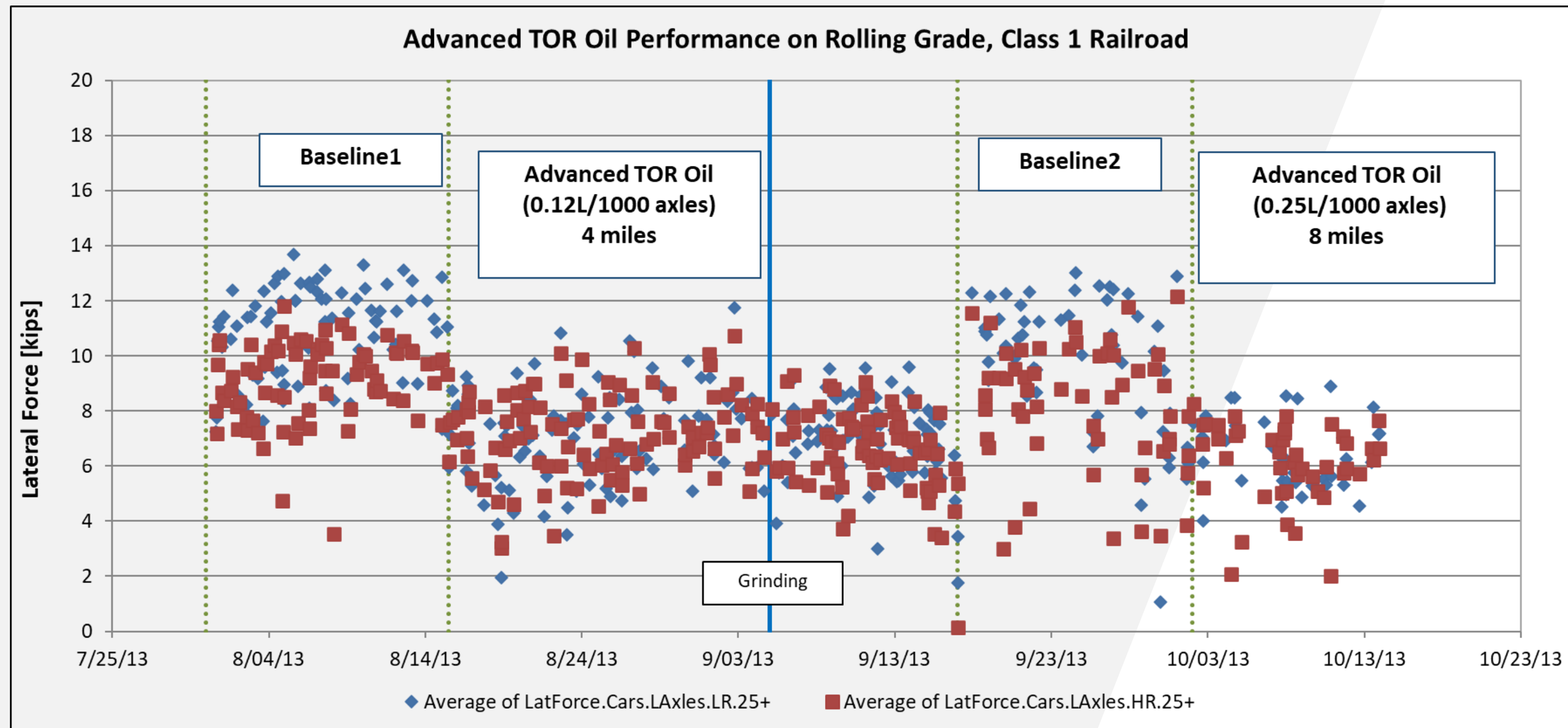
ER LT Performance on Rolling Grade
MP 27.3, 7°30' RH, Main Track, Avg. Lat. Force Distribution, ISI Crib 1, Westbound, High Rail,
Leading Axles, Wheel Load > 25+ kips 24-28 mph speeds, axles ≥ 350





Friction Modifiers Results

Example of L/V data for TOR application – average forces of leading axles per train





Lateral Force Measurement Techniques

The most common methods of measuring lateral forces:

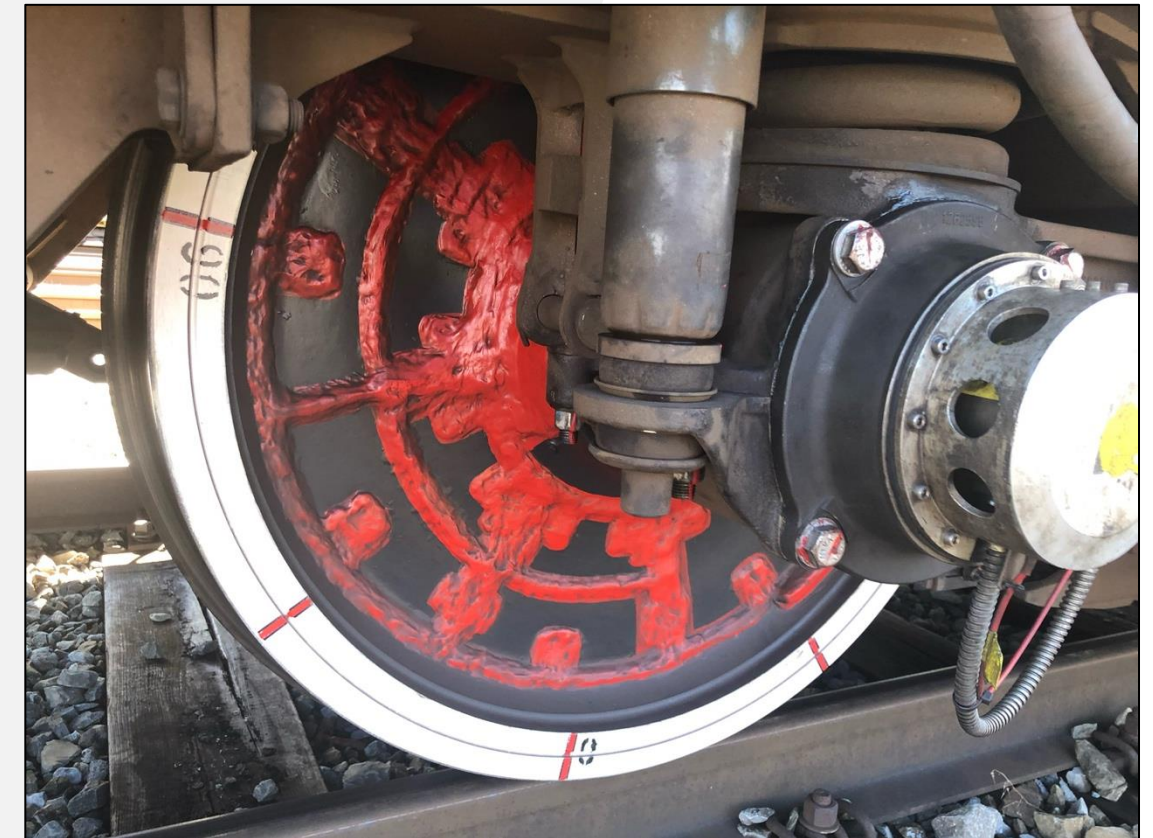
INSTRUMENTED WHEELSETS:

- Use strain gauges mounted directly to the wheel body
- Requires significant modification to wheels/axle
- Measure vertical and lateral forces simultaneously as the train moves

STRAIN GAUGES

- Strain Gauges – installed directly to the rail web to measure track loading
- Micro-welded to the rail between ties, measures bending moment

Class 1s mainly utilize track-based strain gauges for the validation of friction management products





Strain Gauge Based Measurement

- Each site typically has 8 gauges micro-welded to the web and foot of the rail
- Connected by a series of wires/cables to data collection system located at the trackside

Operational Difficulties:

- Installation Precision – sensors must be placed and welded/bonded to the rails in an exact manner. Requiring time, highly trained technicians and complex calibration procedure
- Limited Lifespan – strain gauges and the cabling are delicate and require regular replacement. Harsh operating conditions and due to regular track maintenance activity and equipment
- Temperature Effects – rail material properties and the gauges are affected by temperature change, causing drift of data



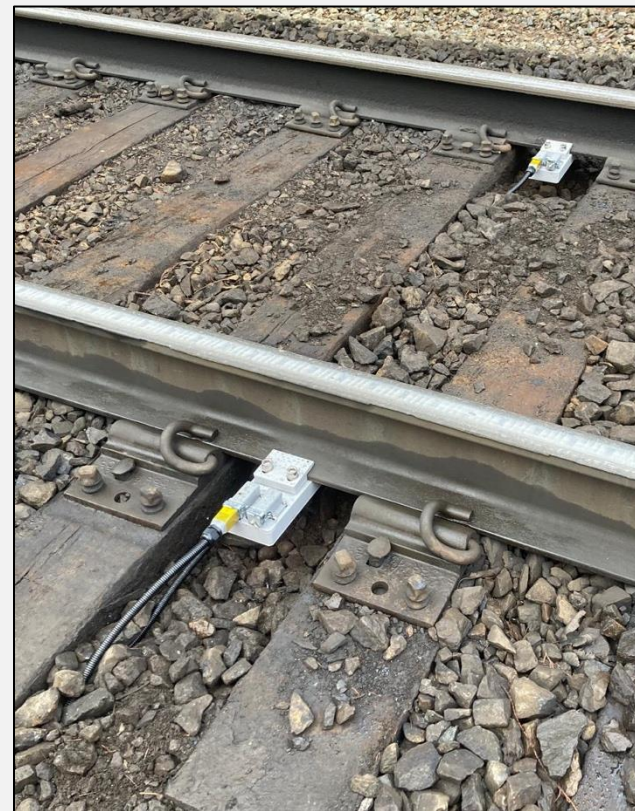
DALLAS, TX



An Ultrasonic Approach

A rapidly deployable under-rail clamp system characterizes vertical and lateral load of every passing wheelset. The units can easily be moved between target curves, providing real-time empirical FM performance data to support LB Fosters total friction management service.

A portable battery-powered datalogger is mounted wayside capturing data locally with optional cloud reporting. For installations lasting more than 1 week, a solar-panel or mains power is required.



DALLAS, TX



The Equipment

Installation takes less than 10 minutes enabling safe installation without disruption to train scheduling.

Installation requires only basic hand tools and very little training, enabling this technology to be rolled out by the existing railroad workforce.



Under-Rail Clamp Installation





How it works?

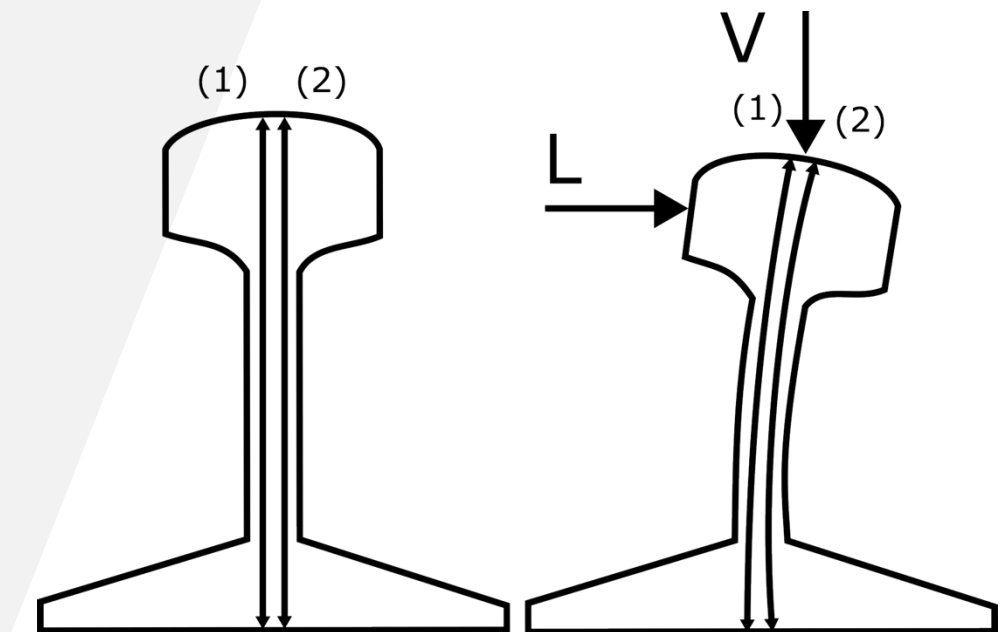
The instrumented clamps are bolted to the underside of the rail foot and can be installed on either the high rail, low rail, or both.

Proprietary ultrasonic sensor arrays are integrated within the clamps that send soundwaves up through the web and reflect off the rail head.

The rate of the ultrasonic wave propagation is proportional to the magnitude and direction of the stress; an established phenomenon known as the acoustoelastic effect (commonly used in automotive engines).

The time it takes for these sound waves to be received by the same ultrasonic sensors can be directly related to the vertical and lateral load.

The reflected waves can be stored locally for post-analysis, and/or they can be edge processed and exported, enabling near real time monitoring of how a particular friction modifier is performing and its impact on track health.





Field Test Site Overview

Field testing was conducted to look at the system's ability to measure lateral force and general usability in track.

- Class 1 track
- 50MGT, directional traffic
- Mainly Unloaded coal/grain, intermodal and mixed freight
- Curve 5°42' left hand
- 0.9% ascending grade
- Speed 30mph
- Wooden tie construction
- GF lubrication good
- TOR unit 2.5 miles from curve





Test Schedule and Plan

The ultrasonic clamps were installed over a period of three days, and the traffic was monitored over this period.

The first day was conducted with no friction modifier applied. The friction modifier applicator was switched on for the second day.

Analysis of two similar empty grain trains that passed on separate days is presented, where friction modifier was only applied on the second day. This will support the assessment of the friction modifier performance.

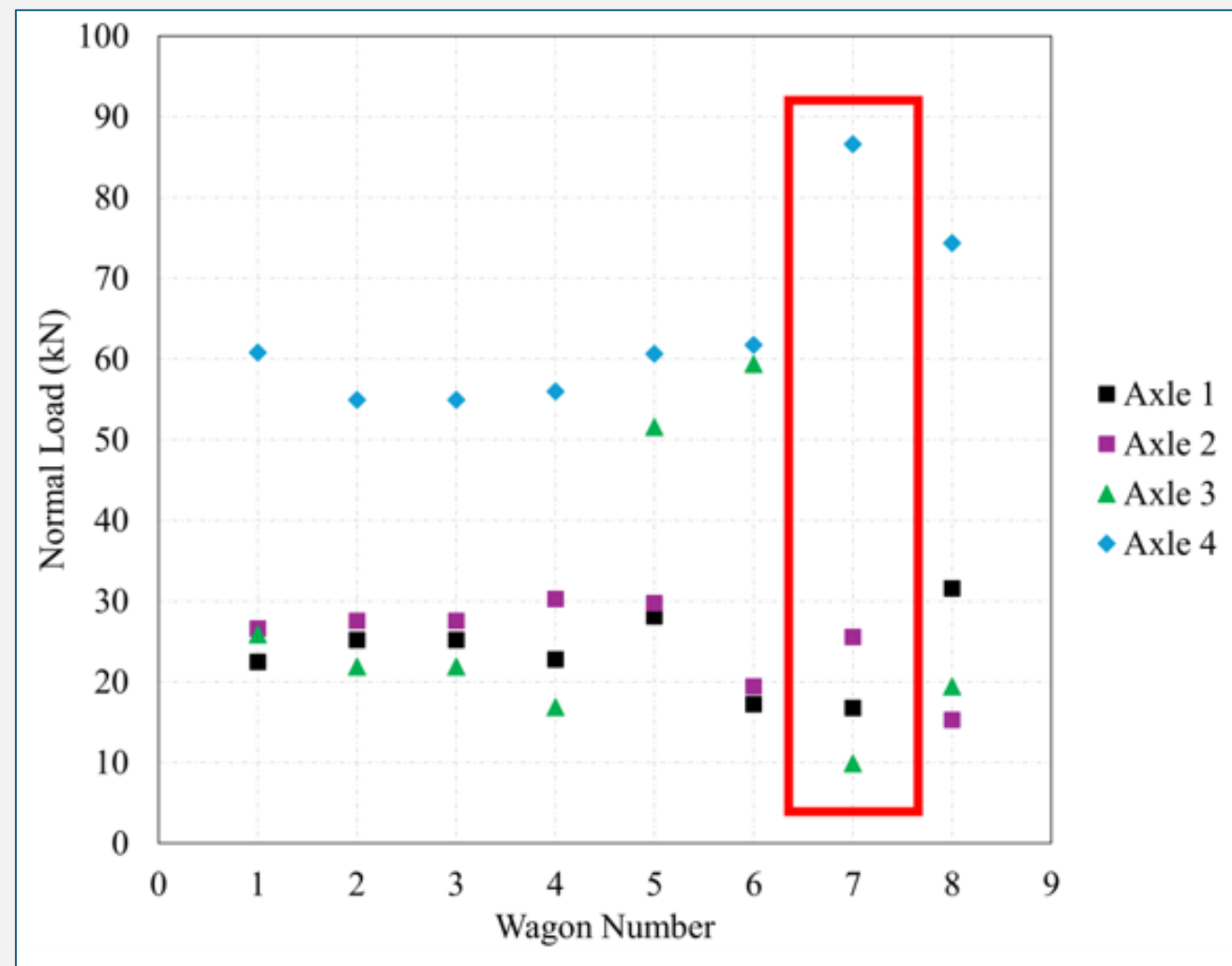




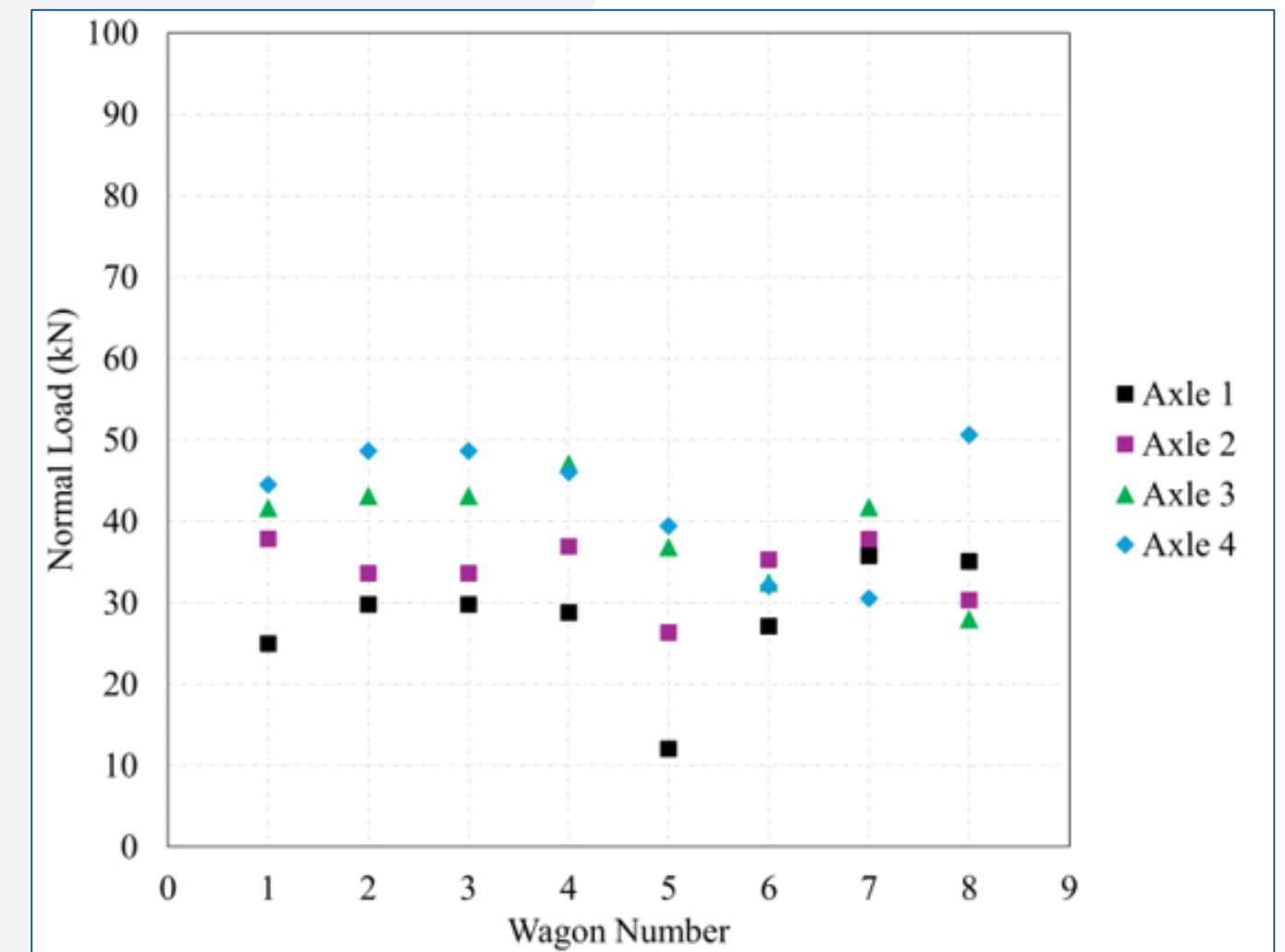
Normal Load Results

The plot and table shows the measured vertical/normal load for the 4 axles of 8 similar wagons. This demonstrates the equipment's capacity to identify outlier wheelsets and axle load distribution.

Low rail loading



High rail loading





Lateral Load Results

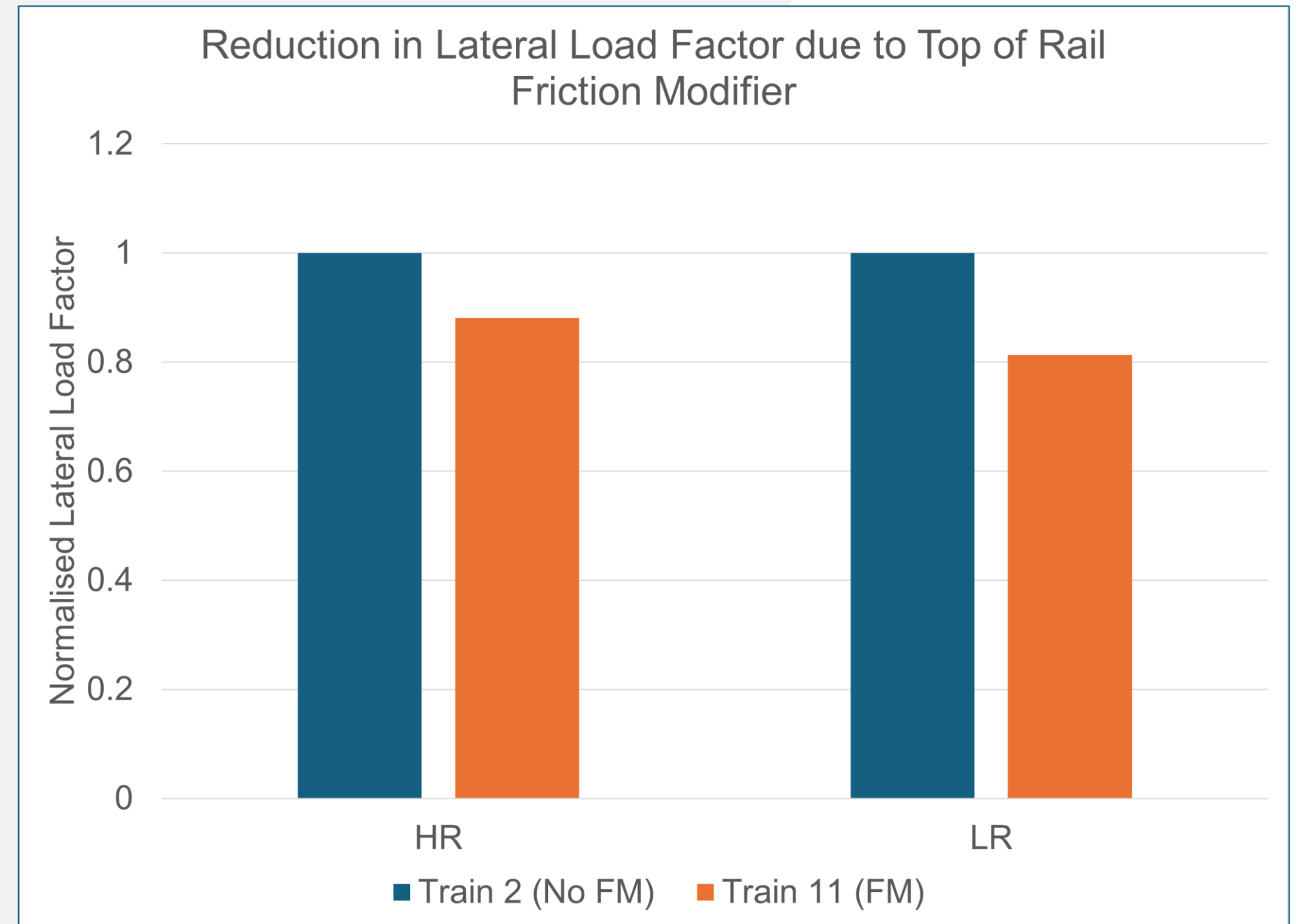
The plot shows the averaged measured lateral load for two similar grain train passes (both high and low rail plotted):

Train 2: Day 1 – no FM product

Train 11: Day 2 – with FM product

The results are averaged over multiple axles. For this work, a normalised comparative lateral load coefficient is reported.

Both the high-rail and low rail consistently showed a reduction in measured lateral load with the additional of FM product.





Lateral Load Results

We saw lateral load reductions on both rails when a top of rail friction modifier was applied.

We saw a 12% reduction in lateral loading on the high rail and a 19% reduction on the low rail for this curve.

Vehicle	HR	LR
Train 002 (No FM)	1	1
Train 011 (FM)	0.88	0.81
Lateral Load Reduction	12%	19%



Next Steps

- New under-rail clamp set and datalogger delivered to LB Foster to support the Total Friction Management service offering
- Solar powered operation with automated reporting and data access via cloud
- RFID tag reader will be added for traffic identification
- Empirical lateral and vertical load data for each individual axle
- System validation/qualification trials planned with L/V load-cell calibration jack
- 10-week Class 1 test campaign scheduled for June 2026
- Test plan to include multiple curves and river/heavy grade site locations



CONCLUSIONS



- A rapidly deployable under-rail measurement clamp system was successfully implemented on a Class 1 track
- Successfully demonstrated the capability of measuring vertical and lateral loads and quantifying the effectiveness of friction management programs
- Future tests are planned which aim to both link these measurements to force and run for longer test durations





Thank you!

Questions?

